



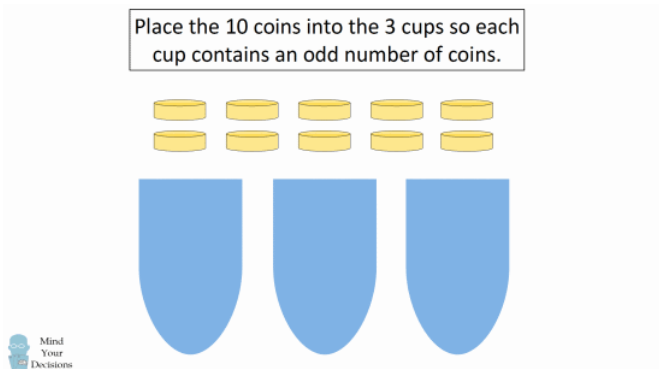
Mind Your Decisions

Presh Talwalkar

10 Coins In 3 Cups – Sunday Puzzle

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Posted November 15, 2015 By Presh Talwalkar. Read [about me](#), or [email me](#).



Can you place 10 coins in 3 cups so that each cup contains an odd number of coins?

This puzzle is sometimes asked as an interview question. It also makes for a fun problem at parties and pubs.

Give it a try and watch the video for a solution.

10 coins and 3 cups riddle

How To Solve The TRICKY 10 Coins, 3 Cups Interview Question



Or scroll down for a text/image solution.

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Answer To 10 Coins In 3 Cups

There are only a limited number of ways that 3 positive numbers can add up to 10.

- 1 + 1 + 8
- 1 + 2 + 7
- 1 + 3 + 6
- 1 + 4 + 5
- 2 + 2 + 6
- 2 + 3 + 5
- 2 + 4 + 4
- 3 + 3 + 4

Notice that in each case at least one of the numbers is even. So it is not possible to place odd coins in each of the 3 cups.

Or think about it another way. If the cups have odd numbers $2x + 1$, $2y + 1$, and $2z + 1$, then for the total to be 10, we must have:

$(2x + 1) + (2y + 1) + (2z + 1) = 10$

Which means:

$2(x + y + z) = 7$

We cannot have 2 times a positive number equal 7, so this is impossible.

So we need to be creative. Consider the partition $2 + 3 + 5$.

Then place the middle cup with 3 INSIDE the cup with 2 coins.

Now each cup contains an odd number of coins!

In fact this solution method works for other partitions that have 1 even number and 2 odd numbers. Place the coins according to the partition, and then place one of the odd cups inside the even cup.

So we get solutions for the cases:

- 1 + 1 + 8
- 1 + 2 + 7
- 1 + 3 + 6
- 1 + 4 + 5
- 2 + 3 + 5
- 3 + 3 + 4

We cannot do this if the partition has only even numbers:

- 2 + 2 + 6 (can't be done)
- 2 + 4 + 4 (can't be done)

We can also place 0 coins in a cup to get a few more solutions.

- 0, 1, 9
- 0, 3, 7
- 0, 5, 5

Place one of the cups with an odd number of coins into the 0 cup, and once again all 3 cups will have an odd number of coins.

PUBLISHED BY

PRESH TALWALKAR

I run the MindYourDecisions channel on [YouTube](#), which has over 1 million subscribers and 200 million views. I am also the author of [The Joy of Game Theory: An Introduction to Strategic Thinking](#), and several other books which are available on [Amazon](#).

(As you might expect, the links for my books go to their listings on Amazon. As an Amazon Associate I earn from qualifying purchases. This does not affect the price you pay.)

By way of history, I started the Mind Your Decisions blog back in 2007 to share a bit of math, personal finance, personal thoughts, and game theory. It's been quite a journey! I thank everyone that has shared my work, and I am very grateful for coverage in the [press](#), including the Shorty Awards, The Telegraph, Freakonomics, and many other popular outlets.

I studied Economics and Mathematics at Stanford University.

People often ask how I make the videos. Like many YouTubers I use popular software to prepare my videos. You can search for animation software tutorials on YouTube to learn how to make videos. Be prepared—animation is time consuming and software can be expensive!

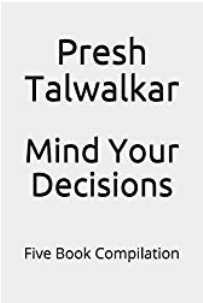
Feel free to send me an email presh@mindyourdecisions.com. I get so many emails that I may not reply, but I save all suggestions for puzzles/video topics.

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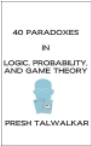


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3 thoughts on “10 Coins In 3 Cups – Sunday Puzzle”

.mau.

15 November 2015 at 7:16

you may also have 1,1,8 coins in the cups. The first two cups have the same number of coins, while the third one is “odd” :-)

Presh Talwalkar

16 November 2015 at 1:45

Clever ;)

Boboccio

16 November 2015 at 6:32

That is a genius trick. I will use this at a party for sure.

Thanks for posting the text solution too, alongside the video.

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